SRI MOHANA PRIYAN

GEN-AI ENGINEER

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My Portfolio
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SUMMARY

I'm a self-taught AI Engineer with a keen eye for product design. I passionately bridge the gap between complex machine learning algorithms and intuitive user experiences. My goal is to develop high-impact AI solutions from computer vision to LLMs that are not only technically powerful but also accessible and user-centric.

SKILLS

- **LANGUAGE:** Python, SQL.
- **FRAMEWORKS & LIBRARIES: GenAI& RAG**: LangChain, LangGraph, Gemini API, ChromaDB. **Deep Learning**: Tensorflow, Keras, Pytorch, OpenCV. **Machine Learning**: Scikit-learn, Seaborn, Tableau, Pandas, Numpy, XGBoost. **Deployment**: Docker, Google Cloud Run, Streamlit.
- **CORE COMPETENCIES:** Generative AI, RAG, AI Agents, Machine Learning, Deep Learning, NLP, Data Analysis.
- **TOOLS & CLOUD:** Google Cloud Platform (GCP), Google Looker Studio, Git, GitHub, Docker.
- **IDEs:** VS Code, Jupyter Notebook, Google Colab, Spyder.

PROJECTS

AUTONOMOUS HYBRID RAG AGENT

Python, LangChain, Google Gemini Pro, ChromaDB, Docker, GCP, Streamlit.

- Engineered an **Agentic RAG** system using **LangGraph** and **Gemini 1.5**, featuring a cyclic state machine that autonomously switches between internal document retrieval and real-time **Web Search** to answer complex queries.
- Implemented a **Self-Correction** workflow with a custom **LLM** as-a-Judge node that evaluates document relevance (Grader), automatically rewriting search queries or triggering web fallbacks to reduce hallucinations by 40%+.
- Developed a **Hybrid Information Retrieval** pipeline using **ChromaDB** for vector storage and **Tavily API** for external context, ensuring high accuracy by prioritising private data before accessing public knowledge.
- Deployed a full-stack interface on **Streamlit Cloud**, utilizing Session State to visualise the agent's decision-making process (e.g., "Grading Documents," "Rewriting Query") for improved user trust and interpretability.

NEXUS BROWSER

Python, LangGraph, Google Gemini Pro, Tavily API, Docker, GCP, RAG.

- Architected an autonomous **RAG** search engine using **FastAPI** and Next.js, integrating Google Gemini Pro and **Tavily API** to deliver real-time, citation-backed answers.
- Engineered a **Generative UI** system that dynamically transforms raw **LLM** JSON streams into interactive **Recharts visualisations** and comparison tables on the fly.
- Developed a stateful **AI agent** with **LangGraph**, enabling cyclic reasoning workflows to refine search queries and improve result accuracy autonomously.
- Deployed a **scalable microservices** architecture on **Google Cloud Run** using **Docker**, utilising auto-scaling to efficiently handle production traffic.
- Solved complex production challenges, including cross-origin (CORS) security and infrastructure configuration, ensuring a seamless, low-latency user experience.

TELECOM CHURN PREDECTOR

Python, XGBoost, Streamlit, SHAP (Explainable AI), Google Gemini Pro, GCP.

- Built an end-to-end retention system that predicts customer churn probability and auto-generates personalised retention strategies using **Generative AI.**
- Developed a high-performance **XGBoost** Classifier to predict churn risk based on tenure, contract type, and monthly charges.
- Integrated **SHAP (Shapley Additive Explanations)** to provide real-time interpretability, explaining why a specific customer is at risk
- Engineered a **GenAI** Feedback Loop using **Google Gemini** that interprets risk factors and drafts personalised retention emails/offers instantly.
- **Deployment:** Containerised with **Docker** and deployed to **Google Cloud Run** with a **CI/CD**
- style workflow.

PNEUMONIA DETECTOR FROM X-RAYS

Python, TensorFlow/Keras, CNN, Streamlit, Docker, GCP.

- Developed a medical diagnostic tool that detects pneumonia from Chest X-Ray images using **Deep Learning**, deployed as a scalable web application.
- Designed and trained a **Convolutional Neural Network (CNN)** on the Chest X-Ray Images (Pneumonia) dataset, achieving high accuracy in classifying Normal vs. Pneumonia cases.
- Implemented data augmentation techniques (rescaling, shear, zoom) to prevent overfitting and improve model generalisation.
- Built an interactive frontend using **Streamlit** that allows doctors/users to upload X-Rays and get instant diagnostic predictions with confidence scores.
- **Deployment:** Dockerized the application and deployed it to **Google Cloud Run**, enabling serverless, auto-scaling access.

EDUCATION

Bachelor’s Degree in Electronics and Communication Engineering **2020 - 2024**

Kongunadu College of Engineering and Technology, Namakkal.

CERTIFICATION

Prompt Design in Vertex AI Skill Badge Offered by **Google** on July-07-2025